

```
from keras.models import Sequential
from keras.layers import Dense

model = Sequential()

model.add(Dense(4, input_shape=(3,), activation='relu'))

model.add(Dense(4, activation='relu'))
model.add(Dense(4, activation='relu'))
model.add(Dense(4, activation='relu'))

model.add(Dense(3, activation='softmax'))

model.compile(optimizer='sgd', loss='categorical_crossentropy', metrics=['accuracy'])

model.summary()

history = model.fit(X_train, y_train, epochs=100, batch_size=10, validation_split=0.2)
```